

Stanley College of Engineering & Technology for Women (A)**B.E (CSE/CME/EEE) IV Semester (Main) Examinations -July-2023****Effective Technical Communication****Time: 03 Hours****Max. Mar****71**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Explain any TWO barriers for communication. [2M CO1 BTL 1]
- b. What is the process of communication? [2M CO2 BTL1]
- c. Explain any TWO types of listening skills. [2M CO3 BTL1]
- d. What is kinesics? [2M CO4 BTL2]
- e. Write about the purpose of user manual. [2M CO5 BTL2]
- f. Write advantages of an Email. [2M CO4 BTL3]

Part-B**4X12=48M**

2. a) Explain various barriers for technical communication. [6M CO1 BTL2]
- b) Discuss various techniques for effective presentation. [6M CO2 BTL3]
3. a) What are verbal and non-verbal communications? Mention some persuasive techniques. [6M CO2BTL3]
- b) Write a user manual for a smart watch. [6M CO3 BTL3]
4. a) What is mobile etiquette? Explain in detail. [6M CO4 BTL1]
- b) Define "Agenda" and "Minutes of meeting." Write the purpose of Agenda and Minutes of Meeting. [6M CO4 BTL3]
5. a) Imagine that you purchased an LG Top Load washing machine from online platform Amazon. You received a damaged product. Write a complaint letter to the Customer Care, Amazon. [6M CO4 BTL6]
- b) Who writes and reads reports? Explain the elements of effective business reports. [6M CO4 BTL6]
6. a) You would like to open a startup business after your graduation. Prepare your business proposal for the same. [6M CO4 BTL6]
- b) What is report? Write various types of a report in detail. [6M CO5 BTL2]
7. a) Explain the barriers for effective listening. [6M CO2 BTL2]
- b) What does 'listening mean'? Explain Active and Passive listening and advantages of listening. [6M CO2 BTL2]

EEE

Hall Ticket No. -

Code: R123746

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) IV Semester (Main) Examinations-July-2023

Electrical Machines-I

Time: 03 Hours

Max. Marks-60

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Part-A**6X2=12M**

- Explain the processes of Electro Mechanical energy conversion. [2M CO1 BTL2]
 - Define the terms of critical resistance and critical speed. [2M CO2 BTL1]
 - What is the back e.m.f or counter voltage? [2M CO3 BTL1]
 - Define voltage regulation of a transformer. [2M CO4 BTL1]
 - What is the condition for maximum efficiency of a transformer? [2M CO5 BTL1]
 - What happens when a DC supply is applied to a Transformer?" [2M CO1 BTL1]

Part-B**4X12=48M**

- Derive the expressions for singly excited systems. [6M CO1 BTL6]
 - Write a short note on energy balance equation in motor. [6M CO1 BTL1]
- Explain how the voltage is developed in a self-excited D.C shunt generator? [6M CO2 BTL2]
 - Draw the construction of DC generator and explain its parts briefly. [6M CO2 BTL6]
- Explain the Speed control of d.c. Motors using Armature voltage and field flux control methods. [6M CO3 BTL2]
 - Describe the brake test on DC motor to determine the performance characteristics of DC motor. [6M CO3 BTL1]
- Why transformer rating is in kVA not in kW? A 230 V, 2.5 kVA single -phase transformer has an iron loss of 100W at 40-Hz and 70 W at 30-Hz. Find the hysteresis and eddy current losses at 50 -Hz. [8M CO4 BTL1]
 - Derive the condition for maximum efficiency in transformer. [4M CO4 BTL6]
- Mention the different types of three phase transformer connection and explain them briefly with neat diagrams. [6M CO5 BTL3]
 - What is an auto-transformer? What advantages are possessed by auto-transformers? [6M CO5 BTL1]
- What will happen to the speed of DC motor when its flux approaches zero? [4M CO3 BTL1]
 - Draw the circuit model of various types of DC motors. A 220-V d.c machine has an armature resistance of 0.5Ω . If the full-load armature current is 20 A, find the induced e.m.f. when the machine acts as a motor. [8M CO3 BTL1]



Hall Ticket No. -

Code: R123751

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) IV Semester (Main) Examinations-July-2023

Switching Theory and Logic Design

Time: 03 Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. Obtain complement and dual for the given expression $(AB+BC+AC)$ (EF) [2M CO1 BTL6]
- b. Draw the diagram of subtractor using truth tables. [2M CO1 BTL4]
- c. What is race around condition in JK flip-flop? [2M CO3 BTL3]
- d. Write the purpose of parallel comparator A/D converter [2M CO4 BTL1]
- e. Differentiate between EPROM and EEPROM. [2M CO5 BTL2]
- f. How a D-flip flop obtained from JK flip flop? Write its truth table. [2M CO5 BTL1]

Part-B

4X12=48M

1. a) Implement the following function using only NOR gates $F = a \cdot (b + c \cdot d) + (b \cdot c)$. [6M CO1 BTL3]
 b) Implement the following function using only NAND gates $G = (a + b) \cdot (c \cdot d + e)$. [6M CO1 BTL3]
2. a) Design a full-adder with two half-adders and basic gates. [6M CO2 BTL4]
 b) Convert Excess-3 code to BCD using Full adder circuits. [6M CO2 BTL5]
3. a) Differentiate between asynchronous and synchronous sequential circuit. Also design MOD 5 asynchronous counter using JK flip flop. [6M CO3 BTL2]
 b) Draw the logic diagram of a SR latch using NOR gates. Explain its Operation using excitation table. [6M CO3 BTL4]
4. a) Discuss the successive approximation analog to digital converters. [7M CO4 BTL3]
 b) Write the specifications and examples of analog to digital converters [5M CO4 BTL1]
5. a) Design a BCD to Excess-3 code converter using a PROM. [7M CO5 BTL6]
 b) Give the comparison between PROM, PLA and PAL. [5M CO5 BTL5]
6. Simplify the following Boolean expressions using K-map and implement it by using NOR gates.
 a) $F(A, B, C, D) = AB'C' + AC + A'CD'$. [6M CO2 BTL4]
 b) $F(W, X, Y, Z) = w'x'y'z' + wxy'z' + w'x'yz + wxyz$. [6M CO2 BTL4]
7. a) Explain M Method of function realization. [7M CO3 BTL2]
 b) Write the difference between combinational circuit and sequential circuit. [5M CO3 BTL2]

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) IV Semester (Main) Examinations-July-2023

Control Systems

Time: 03Hours

Max. Marks-60

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Part-A

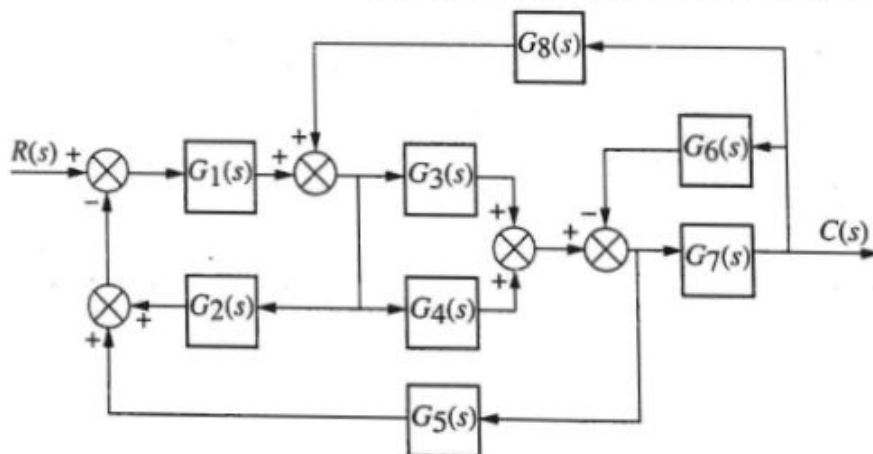
6X2=12M

1. a. List the advantages and disadvantages of open loop system and closed loop system? [2M CO1 BTL1]
- b. Write the analogous electrical elements in force-voltage analogy for the elements of mechanical translational system? [2M CO1 BTL1]
- c. Find the transfer function of the system whose step response is a unit impulse function? [2M CO2 BTL2]
- d. Explain gain margin and phase margin of the control system? [2M CO3 BTL2]
- e. Draw the pole zero plot of lead compensator and lag compensator? [2M CO4 BTL1]
- f. Explain the concept of state? [2M CO5 BTL2]

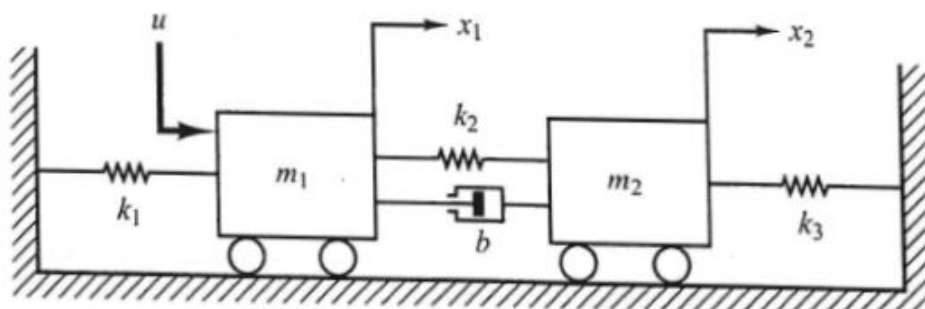
Part-B

4X12=48M

2. a. Find the transfer function $C(s)/R(s)$ for the figure shown below? [6M CO1 BTL2]



- b. Obtain the transfer functions $X_1(s)/U(s)$ and $X_2(s)/U(s)$ of the mechanical system shown in figure below? [6M CO1 BTL2]



3. a. A unity feedback control system has an open loop transfer function $G(s) = \frac{10}{s(s+2)}$. Calculate the damping ratio, undamped natural frequency of oscillations of the system, rise time, peak time, settling time and percentage peak overshoot for a unit step input? [6M CO2 BTL3]

- b. What is Step Response? Explain Time Domain Specifications. [6M CO2 BTL1]

4. The open loop transfer function of a control system is given by [12M CO3 BTL4]

$$G(s)H(s) = \frac{K}{s(s+8)(s^2+2s+5)}$$

Sketch the root locus plot and determine, (i) the break-away points, (ii) the angle of departure from complex poles and (iii) condition for stability?

5. a. Comment on the range of phase margin and gain margin for the following stability conditions of the control system [4M CO4 BTL1]

- Unstable
- Marginally stable
- Absolutely stable
- Conditionally stable

- b. Sketch the Bode plot for the following transfer function [8M CO4 BTL4]

$$G(s)H(s) = \frac{8}{s(1+0.3s)(1+0.1s)}$$

Determine the following of the system

- Gain crossover frequency
- Phase crossover frequency
- Gain margin and
- Phase margin

6. Evaluate the observability of the state space representation of the system given below? [12M CO5 BTL3]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -2 & -2 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

7. a. Draw lag-lead compensator and derive its transfer function? [6M CO4 BTL2]

- b. What are the properties of state transition matrix? Compute state transition matrix. [6M CO5 BTL6]

Hall Ticket No. -

Code: R123563

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) IV Semester (Main) Examinations-July-2023

OOPs Using Java

Time: 03 Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. List the benefits of object-oriented development. [2M CO1 BTL1]
b. Define Inheritance. [2M CO2 BTL1]
c. Illustrate the use of string tokenizer. [2M CO3 BTL2]
d. List out AWT classes. [2M CO4 BTL1]
e. Write about Files. [2M CO5 BTL2]
f. Define package. [2M CO2 BTL1]

Part-B

4X12=48M

2. a) Explain in detail object oriented concepts. [6M CO1 BTL5]
b) Write about variables and arrays with example. [6M CO1 BTL2]
3. a) Discuss about classes and methods [6M CO2 BTL6]
b) Explain Exception Handling with example program. [6M CO2 BTL2]
4. a) Describe any three Collection Classes. [6M CO3 BTL2]
b) Explain any three Legacy classes with example. [6M CO3 BTL2]
5. a) Explain in detail Two Event Handling Mechanisms. [6M CO4 BTL2]
b) Discuss about Labels, Buttons and Check Boxes with example program. [6M CO4 BTL6]
6. a) Explain Files with example program in java. [6M CO5 BTL2]
b) Explain the Serialization and Deserialization concepts in detail. [6M CO5 BTL2]
7. a) List out and explain Layout managers in AWT. [6M CO4 BTL1]
b) Explain Operators in detail. [6M CO1 BTL2]

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